Real-Time Human Violence Detection Using ResNet3D-18 with Confidence-Gated Temporal Verification

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***Abstract*— Automated solutions that can identify violent activity in real time are desperately needed due to the surveillance infrastructure's increasing growth. Continuous video streams are ineffective and prone to human mistake when monitored by hand, especially in settings with extensive surveillance. The ResNet3D-18 architecture is used in this paper to propose a real-time human violence detection system based on a 3D Convolutional Neural Network. In order to identify violent and non-violent behaviors, the suggested method analyzes brief video clips of sixteen consecutive frames to identify spatiotemporal motion patterns. A confidence-gated temporal verification mechanism is proposed to improve detection stability and decrease false positives. This mechanism requires continuous high-confidence predictions prior to generating an alert. Accuracy, precision, recall, and F1-score are among the common classification metrics used to assess the model. In both CPU and GPU contexts, experimental results show dependable real-time performance and efficient spatiotemporal feature learning. By offering a computationally effective and deployment-ready solution for intelligent surveillance systems, the suggested method enhances public safety monitoring.**

Keywords— Violence Detection, 3D CNN, Spatio-Temporal Modeling, Deep Learning, Confidence-Gated Verification, Intelligent Surveillance Systems

I. INTRODUCTION

Due to the widespread use of surveillance cameras in both public and private settings, there is an excessive amount of video data that human operators are unable to effectively monitor. Observing video feeds by hand is time-consuming, prone to mistakes, and frequently causes delays in responding to significant incidents, especially in busy or extensive deployments. As a result, in recent years, the need for automated violence detection systems has increased significantly [1].

With designs like three-dimensional convolutional neural networks (3D CNNs) showing higher performance by simultaneously learning both spatial and temporal data, deep learning has become a potent tool for action recognition in videos [2], [4]. When applied to dynamic behaviors like violent actions, traditional 2D CNN techniques have limitations since they are unable to capture temporal motion effectively. Large-scale datasets like RWF-2000, which offer realistic surveillance scenarios for model validation, have sped up research in real-world violence detection [1].

In order to enhance long-range temporal modeling in video classification tasks, more recent research has looked into transformer-based models and attention mechanisms [2], [3]. In real-world surveillance applications, solid real-time performance and dependable alarm mechanisms are crucial, according to surveys on violence detection [11].

Despite these developments, a lot of current systems are either unstable in real-time or generate a lot of false positives because they rely on instantaneous forecasts. In order to overcome these difficulties, this study suggests a ResNet3D-18-based real-time violence detection system with a confidence-gated temporal verification mechanism that increases robustness and lowers false alarms in real-world deployment scenarios.

II. RELATED WORK

Automated violence detection in surveillance videos has attracted significant research attention due to increasing security demands in public and private environments. Early approaches primarily relied on handcrafted motion descriptors and traditional machine learning classifiers; however, these methods struggled to generalize under complex real-world conditions such as occlusion, lighting variation, and crowd density.

The introduction of large-scale annotated datasets, particularly RWF-2000 [1], enabled the development and benchmarking of deep learning-based violence detection systems in realistic surveillance settings. The availability of such datasets accelerated progress toward models capable of handling diverse environmental scenarios.

Recent research has largely shifted toward deep spatio-temporal architectures. Three-dimensional convolutional neural networks (3D CNNs) extend conventional 2D convolution by incorporating a temporal dimension into the filtering process, allowing simultaneous modeling of motion and appearance information [2], [4]. These architectures have demonstrated strong performance in video action recognition tasks where short-term motion cues are critical.

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In addition to convolution-based methods, transformer-based video models such as Video Swin Transformer have been proposed to capture long-range temporal dependencies through self-attention mechanisms [2], [3]. Although the accuracy of these architectures is competitive, their computational complexity frequently prevents them from being used in real-time surveillance settings.

The goal of several recent studies has been to increase deployment feasibility through the creation of multi-stream frameworks or lightweight architectures specifically suited for real-time applications [5], [12]. Other methods improve robustness in crowded scenes through using skeleton-based representations or anomaly detection methods [7], [9]. Many present ones rely on frame-level or single-clip predictions, which can result in unstable alerts under dynamic conditions, despite its requiring classification performance.

III. PROPOSED METHODOLOGY

The system that is suggested is to detect violence in real time from surveillance video recordings using spatiotemporal deep learning. Video acquisition, preprocessing, 3D feature extraction, classification, and temporal verification based on confidence make up the overall system.

A. System Overview

From a camera or security feed, the system continuously records video input. A sliding window technique gathers 16 consecutive frames to create a brief video clip rather than processing each frame separately. Effective modeling of motion dynamics is made possible by this clip-based method, which is essential for differentiating between violent and non-violent activities.

Before being sent to the neural network, each video clip is normalized and scaled to 112 × 112 pixels. The input tensor that has been processed is shown as:

$$X \in \mathbb{R}^{3\times16\times112\times112} \tag{1}$$

In this case, 3 stands for RGB channels, 16 for temporal frames, and 112 x 112 for spatial resolution.

B. Spatio – Temporal Feature Extraction

The ResNet3D-18 architecture is used by the system in order to capture dynamic motion patterns. When three-dimensional convolutional networks learn both spatial and temporal representations simultaneously, they have shown excellent performance in action recognition [2], [4]. Softmax activation is used to create class probabilities once the last completely connected layer is altered for binary classification (Violent/Non-Violent).

C. Confidence – Gated Temporal Verification

Real-time systems may have unsteady predictions despite the high classification accuracy of deep learning models. This problem is addressed by introducing a confidence level of 0.85. A violent alert is only triggered when five consecutive predictions go above this threshold. This continuous verification near eliminates false positives and increases deployment dependability.

IV. MODEL ARCHITECTURE

For spatiotemporal feature extraction, the proposed method uses a 3D convolutional neural network based on the ResNet3D-18 architecture. To effectively describe motion dynamics in video sequences, three-dimensional CNNs perform convolution operations in both spatial and temporal dimensions. This expands on traditional 2D convolutional networks [2], [4]. As a result, they are well-suited for action recognition and violence detection tasks.

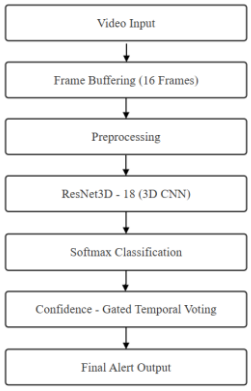

Fig. 1. Proposed system design for real-time violence detection.

A. Backbone Network

Based on the ideas of residual learning, ResNet3D-18 introduces shortcut connections to fix vanishing gradient problems and help with training deeper networks [4]. Using volumetric inputs of size $C \times T \times H \times W$, convolutional kernels in the 3D version enable the simultaneous learning of motion and appearance features.

Batch normalization, ReLU activation algorithms, and stacked residual blocks with 3D convolutional layers make up the network. High-level semantic representations and low-level motion features can be gradually extracted thanks to the hierarchical structure.

B. Classification Layer

Binary categorization (Violent/Non-Violent) is accomplished by modifying the initial fully connected layer of ResNet3D-18. Before the last linear layer, a dropout layer with probability p=0.5 is added to lessen overfitting. 512 extracted features are mapped into two output classes in the last layer. Softmax activation is applied to compute class probabilities:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{2} e^{z_j}} \tag{2}$$

where $z_i$represents the logit score for class i.

C. Architectural Justification

Since 3D CNN-based architectures can train compact spatio-temporal representations, recent studies show their effectiveness for real-time video classification problems [2], [5]. In contrast to transformer-based video models, which often need more processing power [3], 3D CNNs are suitable for surveillance deployment because they provide a good balance between accuracy and real-time performance.

V. MATHEMATICAL FORMULATION

The mathematical representation of the proposed violence detection framework is discussed here. It includes input modeling, the 3D convolution operation, the classification function, and confidence-based temporal verification.

A. Input Representation

Each video clip is shown as a four-dimensional tensor:

$$X \in \mathbb{R}^{C\times T\times H\times W} \tag{3}$$

where H and W refer to spatial dimensions (112 x 112), C=3 indicates RGB channels, and T=16 represents temporal frames. This volumetric representation allows for modeling both spatial and temporal elements simultaneously.

B. 3D Convolution Operation

The main function of the ResNet3D-18 model is the three-dimensional convolution defined as:

$$Y(i,j,k) = \sum_m \sum_n \sum_p X(i+m, j+n, k+p) \cdot W(m,n,p) \quad (4)$$

where $Y(i, j, k)$ is the generated feature map, $X(i, j, k)$ is the input tensor, and W represents learnable 3D filters. Action recognition needs the extraction of motion-aware spatiotemporal representations [2], [4].

C. Softmax Classification

The final layer produces logits z_i, which are converted into probabilities using the softmax function:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{2} e^{z_j}} \quad (5)$$

The predicted class is determined as:

$$\hat{y} = \arg\max_i P(y_i) \quad (6)$$

D. Confidence – Gated Temporal Verification

To improve detection stability, a confidence threshold of $\tau = 0.85$ is used. A prediction is considered valid if:

$$P(\text{Violent}) \geq \tau \quad (7)$$

A final alert occurs only if five consecutive predictions meet the condition.

$$\sum_{i=1}^{5} V_i = 5 \quad (8)$$

where, if the confidence requirement is satisfied, Vi = 1, and otherwise, 0. This continuous verification process improves real-time robustness and lowers false positives.

VI. EXPERIMENTAL RESULTS

The performance of the suggested violence detection approach is assessed in this part utilizing real-time inference analysis and quantitative measurements.

A. Experimental Setup

The proposed ResNet3D-18-based model was trained and evaluated using a dataset focused on detecting violence, which aligns with publicly available benchmarks like RWF-2000 [1]. The dataset was split into training and testing subsets, keeping a balance between violent and non-violent samples.

Each input sample included 16 consecutive frames resized to 112 × 112 resolution. The model used cross-entropy loss optimized through standard backpropagation. To avoid overfitting, the best-performing model was chosen based on validation accuracy.

All experiments took place in both CPU and GPU environments to assess inference efficiency and deployment practicality.

B. Evaluation Metrics

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score.

TABLE I. PERFORMANCE METRICS OF PROPOSED MODEL

Metric	Value (%)
Accuracy	97.95
Precision	98.00
Recall	98.00
F1 - Score	98.00

1. Accuracy
   Accuracy gauges how accurate the classification model is overall:
   $$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$
2. Precision
   The precision metric measures the percentage of correctly predicted violent incidents from all the predicted violent incidents:
   $$\text{Precision} = \frac{TP}{TP+FP} \quad (10)$$
3. Recall
   The recall is the percentage of violent incidents that are correctly identified out of all actual violent incidents. It is also called sensitivity or true positive rate:
   $$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$
4. F1-Score
   The F1-score is the harmonic mean of precision and recall:
   $$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

The high values in all metrics show that the model can clearly tell the difference between violent and non-violent activities.

C. Confusion Matrix Analysis

To gain a better understanding of how the model classifies data, we created a confusion matrix using the test dataset. This matrix summarizes true positives, true negatives, false positives, and false negatives, giving us insight into how well it predicts each class.

The results show a strong ability to distinguish between violent and non-violent categories, with a high number of correctly classified samples. The low number of false positives suggests that the model effectively limits incorrect violence alerts. This is crucial in surveillance systems where reliability matters.

Additionally, the confidence-gated temporal verification mechanism helps improve classification stability. Previous studies highlight the need to reduce false alarms in real-world violence detection systems. By requiring a series of high-confidence predictions before sending alerts, the proposed system becomes more reliable in changing environments.

The analysis of the confusion matrix shows that the framework delivers balanced and dependable performance. It is suitable for real-time deployment.

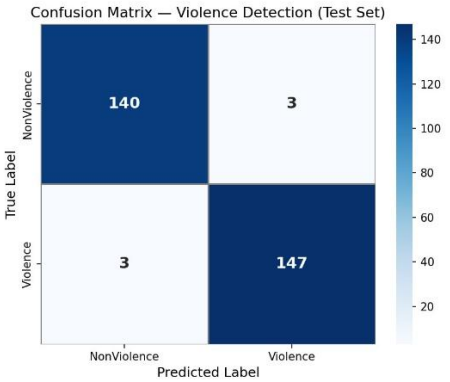

Fig. 2. Confusion matrix of the suggested violence detection model.

D. Real – Time Inference Performance

Real-time feasibility is essential for surveillance applications. We evaluated the proposed framework using both CPU-only and GPU-accelerated setups. GPU-based inference enabled smooth real-time processing, making it suitable for continuous monitoring scenarios. The lightweight design of ResNet3D-18 offered a good balance between computational efficiency and classification performance. The results show that combining 3D spatio-temporal feature extraction with confidence-based temporal verification not only boosts detection accuracy but also enhances operational stability in real-world surveillance deployments.

VII. DISCUSSION

The experimental results show that the ResNet3D-18-based framework effectively captures spatio-temporal motion features related to violent activities in surveillance videos. Three-dimensional convolutional architectures can model spatial and temporal information together, making them a good fit for action recognition tasks [2], [4]. The strong classification performance in this study further confirms the value of 3D CNN-based representations for short-term motion modeling in real-time settings.

Compared to traditional 2D CNN methods that mainly focus on spatial features, 3D CNNs handle consecutive frames as volumetric inputs. This approach allows for better representation of motion. Previous studies have shown that these architectures perform better in recognizing dynamic activities [4], [5]. The results of this work match those findings, especially regarding classification stability and fewer misclassifications of non-violent activities.

The confusion matrix analysis shows a good balance in distinguishing between violent and non-violent classes. The high precision and recall values indicate that the model effectively balances detection sensitivity with reducing false alarms. This balance is especially crucial in surveillance applications, where too many false positives can harm system credibility and operational efficiency, as noted in recent violence detection surveys [11].

A key feature of this framework is the integration of a confidence-gated temporal verification mechanism. While many current systems depend on immediate predictions [6], the proposed approach adds ongoing confidence checks over consecutive clips before sending alerts. This design improves reliability in challenging surveillance situations, where sudden movements, obstructions, or background noise could otherwise cause unstable predictions.

VIII. CONCLUSION

A real-time violence detection framework based on the ResNet3D-18 architecture, which includes a confidence-gated temporal verification approach, was introduced in this paper. The proposed method effectively captures the spatiotemporal motion patterns needed to distinguish between violent and non-violent activities in surveillance footage by using 3D convolutional neural networks.

By reducing false-positive alerts caused by noise or brief motion, continuous confidence-based verification significantly improves detection stability. Experimental evaluations demonstrate strong classification performance across common parameters and maintain real-time viability on realistic deployment platforms.

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